Explicit convergence certificates for Chambolle–Pock at small extrapolation

Abstract

We give explicit convergence certificates for two small extrapolation parameters of the unmodified Chambolle–Pock method. At $\theta = 1/4$, an integer quadratic energy on three consecutive states proves weak convergence whenever $\tau\sigma\|L\|^2 \le 5/2$. The certificate applies to arbitrary bounded linear coupling between real Hilbert spaces, proper lower-semicontinuous convex functions, and arbitrary initial points. It also gives square-summable increments and a summable saddle gap. Its positivity uses the actual proximal history, and all matrix inequalities are certified by exact rational arithmetic. At $\theta = 1/2$, a finite-window argument completes the weak-convergence consequence of a previously published energy throughout $\tau\sigma\|L\|^2 < 2$. This latter range meets the known bilinear obstruction. The quarter-extrapolation bound is not asserted to be optimal.

Keywords: Chambolle–Pock method; primal–dual splitting; Lyapunov functions; weak convergence; exact certificates

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1 Setting and the two parameter choices

Let H, G be real Hilbert spaces, let $f : H \to \mathbb{R} \cup \{+\infty\}$ and $g : G \to \mathbb{R} \cup \{+\infty\}$ be proper lower-semicontinuous convex functions, and let $L : H \to G$ be bounded linear. Assume that the KKT set

$$Z = \{(x^*, y^*) : -L^*y^* \in \partial f(x^*), \quad Lx^* \in \partial g^*(y^*)\} \tag{1}$$

is nonempty. For fixed $\tau, \sigma > 0$ and arbitrary $(x_0, y_0) \in H \times G$, consider

$$x_{k+1} = \operatorname{prox}_{\tau f}(x_k - \tau L^* y_k), \tag{2}$$

$$y_{k+1} = \operatorname{prox}_{\sigma g^*}\big(y_k + \sigma L[x_{k+1} + \theta(x_{k+1} - x_k)]\big). \tag{3}$$

There is no additional relaxation, correction, or change of parameters.

The public preprint of Upadhyaya [1, Theorem 1.2] proves weak convergence for $0 < \theta \le 1/2$ under

$$\tau\sigma\|L\|^2 < \frac{4\theta(2-\theta)}{1 - 2\theta + 9\theta^2 - 4\theta^3}. \tag{4}$$

The right side is $7/4$ at $\theta = 1/4$ and $12/7$ at $\theta = 1/2$. Section 5 of that preprint asks for explicit longer-history certificates, motivated by the numerical Lyapunov regions of AutoLyap [3]. The final article of Banert, Upadhyaya and Giselsson [2, Theorem 2] proves weak convergence for $\theta > 1/2$ and $\tau\sigma\|L\|^2 < 4/(1 + 2\theta)$. Its Lemma 6 also supplies the energy used in our half-extrapolation argument.

Theorem 1. *In the setting above, every sequence generated by* (2)–(3) *converges weakly to a point of* Z *under either of the following conditions:*

$$\theta = \frac{1}{4}, \quad \tau\sigma\|L\|^2 \le \frac{5}{2}; \qquad \text{or} \qquad \theta = \frac{1}{2}, \quad \tau\sigma\|L\|^2 < 2. \tag{5}$$

If H *and* G *are finite dimensional, convergence is in norm.*

The first statement follows from the exact history certificate in Sections 2–4. The second is a short endpoint consequence of the published energy, proved in Section 5. The bilinear necessary boundary at $\theta = 1/4$ is $8/3$ [2, Section 7]; hence the first bound in (5) is an explicit enlargement of (4), not a sharp-threshold claim. No interval of extrapolation parameters is inferred from these two fixed choices.

2 An integer energy at quarter extrapolation

Fix a point $(x^*, y^*) \in Z$. Introduce the scaled variables and coupling

$$X_k = \frac{x_k - x^*}{\sqrt{\tau}}, \qquad Y_k = \frac{y_k - y^*}{\sqrt{\sigma}}, \qquad K = \sqrt{\tau\sigma}\, L, \tag{6}$$

and the nonnegative convex functions

$$\phi(X) = f(x^* + \sqrt{\tau}X) - f(x^*) + \sqrt{\tau}\langle L^* y^*, X\rangle, \tag{7}$$

$$\psi(Y) = g^*(y^* + \sqrt{\sigma}Y) - g^*(y^*) - \sqrt{\sigma}\langle Lx^*, Y\rangle. \tag{8}$$

Both functions vanish and are minimized at zero. Direct substitution in the proximal minimization problems gives, when $\theta = 1/4$,

$$X_{k+1} = \operatorname{prox}_{\phi}(X_k - K^* Y_k), \qquad Y_{k+1} = \operatorname{prox}_{\psi}\big(Y_k + K(5X_{k+1} - X_k)/4\big). \tag{9}$$

All function values at indices $k \ge 1$ are finite, even if the initial pair lies outside the function domains.

For a symmetric real matrix $A = (A_{ij})$ and a vector list $h = (h_i)$ in one Hilbert space, write

$$[A; h] = \sum_{i,j} A_{ij}\langle h_i, h_j\rangle. \tag{10}$$

For $k \ge 2$, define

$$\xi_k = (X_{k-2}, X_{k-1}, X_k, K^*Y_{k-2}, K^*Y_{k-1}, K^*Y_k), \tag{11}$$

$$\eta_k = (Y_{k-2}, Y_{k-1}, Y_k, KX_{k-2}, KX_{k-1}, KX_k), \tag{12}$$

$$E_k = [P; \xi_k] + [Q; \eta_k], \tag{13}$$

where the following matrices have integer entries:

$$P = \begin{pmatrix} 14688 & 14752 & -26464 & 0 & 0 & 0 \\ 14752 & 18976 & 2896 & 0 & 0 & 0 \\ -26464 & 2896 & 49904 & 0 & 0 & 0 \\ 0 & 0 & 0 & 848 & 3744 & -4592 \\ 0 & 0 & 0 & 3744 & 2896 & -6640 \\ 0 & 0 & 0 & -4592 & -6640 & 11232 \end{pmatrix}. \tag{14}$$

$$Q = \begin{pmatrix} 32456 & 15556 & -28832 & -12083 & -11277 & 20384 \\ 15556 & 24976 & 6160 & -984 & 7200 & -384 \\ -28832 & 6160 & 22736 & 8272 & 11520 & -19936 \\ -12083 & -984 & 8272 & -784 & -880 & 1664 \\ -11277 & 7200 & 11520 & -880 & 9216 & -8336 \\ 20384 & -384 & -19936 & 1664 & -8336 & 6672 \end{pmatrix}. \tag{15}$$

The separate matrices need not be positive semidefinite. Positivity holds on histories generated by (9).

Lemma 1. *Suppose $\|K\|^2 \le 5/2$. For every actual trajectory, every KKT point, and every $k \ge 2$,*

$$E_k \ge \|X_k\|^2 + \|Y_k\|^2, \tag{16}$$

$$E_k - E_{k+1} \ge 160\big(\|X_{k+1} - X_k\|^2 + \|Y_{k+1} - Y_k\|^2 + \phi(X_{k+1}) + \psi(Y_{k+1})\big). \tag{17}$$

In particular the product endpoint $\|K\|^2 = 5/2$ is included.

3 The exact certificate identities

We prove Lemma 1 by giving all coefficients of two nonnegative decompositions. The verification uses only subgradient inequalities, the operator norm bound, adjoint identities, and finite positive matrices.

At a fixed $k \ge 2$, relabel the states $k-2, k-1, k, k+1$ by $0, 1, 2, 3$. The two eight-entry vector lists are

$$\begin{aligned} h &= (X_0, X_1, X_2, X_3, K^*Y_0, K^*Y_1, K^*Y_2, K^*Y_3), \\ j &= (Y_0, Y_1, Y_2, Y_3, KX_0, KX_1, KX_2, KX_3). \end{aligned} \tag{18}$$

For $i = 1, 2, 3$, proximal optimality gives

$$a_i = X_{i-1} - X_i - K^*Y_{i-1} \in \partial\phi(X_i), \qquad b_i = Y_{i-1} - Y_i + (5KX_i - KX_{i-1})/4 \in \partial\psi(Y_i). \tag{19}$$

Adjoin a root node $*$ with zero point, value and subgradient. For distinct nodes define

$$C^{\phi}_{i\ell} = \phi(X_i) - \phi(X_\ell) - \langle a_\ell, X_i - X_\ell\rangle \ge 0, \qquad C^{\psi}_{i\ell} = \psi(Y_i) - \psi(Y_\ell) - \langle b_\ell, Y_i - Y_\ell\rangle \ge 0. \tag{20}$$

For descent use nodes $(*, 1, 2, 3)$; for coercivity use only $(*, 1, 2)$. The point with local index 0 is used in the subgradients, but its function value is never needed.

Here is a compact specification of the residual matrices. Let $e_0, \dots, e_7$ be the coordinate vectors of $\mathbb{R}^8$, let $\operatorname{sym}(u,v) = (uv^T + vu^T)/2$, and put

$$J_o = (e_0, e_1, e_2, e_4, e_5, e_6)^T, \quad J_n = (e_1, e_2, e_3, e_5, e_6, e_7)^T, \quad D = (e_3 - e_2)(e_3 - e_2)^T, \tag{21}$$

where a displayed transpose turns the list of column vectors into a row-selector matrix. Set

$$\alpha_i = e_{i-1} - e_i - e_{i+3}, \qquad \beta_i = e_{i-1} - e_i + (5e_{i+4} - e_{i+3})/4, \tag{22}$$

with $e_* = \alpha_* = \beta_* = 0$. The Gram coefficient matrices of (20) are

$$A^{\phi}_{i\ell} = -\operatorname{sym}(\alpha_\ell, e_i - e_\ell), \qquad A^{\psi}_{i\ell} = -\operatorname{sym}(\beta_\ell, e_i - e_\ell). \tag{23}$$

For an index list $I \subset \{0,1,2,3\}$, let A_I select its coordinates from $\mathbb{R}^8$ and let B_I select the corresponding coordinates shifted by four. For symmetric U, V and arbitrary W of size $|I|$, define

$$N_H(I;U,V,W) = \tfrac{5}{2}A_I^T U A_I - B_I^T V B_I + \tfrac{1}{2}(A_I^T W B_I + B_I^T W^T A_I), \tag{24}$$

$$N_G(I;U,V,W) = \tfrac{5}{2}A_I^T V A_I - B_I^T U B_I - \tfrac{1}{2}(B_I^T W A_I + A_I^T W^T B_I). \tag{25}$$

If $U, V \succeq 0$, then

$$\begin{aligned}[N_H;h] + [N_G;j] = \tfrac{5}{2}[U;(X_i)_{i\in I}] - [U;(KX_i)_{i\in I}] \\ + \tfrac{5}{2}[V;(Y_i)_{i\in I}] - [V;(K^*Y_i)_{i\in I}] \ge 0.\end{aligned} \tag{26}$$

The W terms cancel by adjointness. The inequality follows from $\|K\|^2 \le 5/2$ by factoring the finite PSD matrices. It is valid for vectors in arbitrary Hilbert spaces.

The integer coefficient tables in Appendix A specify nonnegative weights w^{ϕ}, w^{ψ} and v^{ϕ}, v^{ψ}, as well as two triples of norm multipliers. Denote the norm matrices for $I = (0,1,2,3)$ by N_H^d, N_G^d, and those for $I = (0,1,2)$ by N_H^c, N_G^c. Define

$$S_H = J_o^T P J_o - J_n^T P J_n - 160D - \sum w^{\phi}_{i\ell} A^{\phi}_{i\ell} - N_H^d, \tag{27}$$

$$S_G = J_o^T Q J_o - J_n^T Q J_n - 160D - \sum w^{\psi}_{i\ell} A^{\psi}_{i\ell} - N_G^d, \tag{28}$$

$$T_H = J_o^T P J_o - e_2 e_2^T - \sum v^{\phi}_{i\ell} A^{\phi}_{i\ell} - N_H^c, \tag{29}$$

$$T_G = J_o^T Q J_o - e_2 e_2^T - \sum v^{\psi}_{i\ell} A^{\psi}_{i\ell} - N_G^c. \tag{30}$$

Sums run over the distinct ordered node pairs for the corresponding weights. Direct coefficient addition gives, for either function value list (q_i) with $q_* = 0$,

$$\sum w^{\phi}_{i\ell}(q_i - q_\ell) = \sum w^{\psi}_{i\ell}(q_i - q_\ell) = -160 q_3, \qquad \sum v^{\phi}_{i\ell}(q_i - q_\ell) = \sum v^{\psi}_{i\ell}(q_i - q_\ell) = 0. \tag{31}$$

Therefore the following are exact identities:

$$\begin{aligned}E_k - E_{k+1} - 160\big(\|X_3 - X_2\|^2 + \|Y_3 - Y_2\|^2 + \phi(X_3) + \psi(Y_3)\big) \\ = \sum (w^{\phi}_{i\ell} C^{\phi}_{i\ell} + w^{\psi}_{i\ell} C^{\psi}_{i\ell}) + [N_H^d;h] + [N_G^d;j] + [S_H;h] + [S_G;j],\end{aligned} \tag{32}$$

$$\begin{aligned}E_k - \|X_2\|^2 - \|Y_2\|^2 \\ = \sum (v^{\phi}_{i\ell} C^{\phi}_{i\ell} + v^{\psi}_{i\ell} C^{\psi}_{i\ell}) + [N_H^c;h] + [N_G^c;j] + [T_H;h] + [T_G;j].\end{aligned} \tag{33}$$

To verify positivity, put $R = (I_3, -\mathbf{1})^T \in \mathbb{R}^{4\times 3}$, where the last row is $(-1,-1,-1)$, and $B = \operatorname{diag}(R, R)$. The exact coefficient tables give

$$U_d\mathbf{1} = V_d\mathbf{1} = 0, \qquad S_H \operatorname{diag}(\mathbf{1},\mathbf{1}) = S_G \operatorname{diag}(\mathbf{1},\mathbf{1}) = 0. \tag{34}$$

Here the latter block-diagonal matrix has two columns and eight rows. The future rows and columns $3, 7$ of T_H, T_G are zero. On the complementary subspaces, all eight matrices listed in Table 1 are positive definite. The table records positive integer lower bounds for their successive exact LDL^T pivots. They are obtained from the displayed integer data using the rational recurrence

$$d_i = m_{ii} - \sum_{r<i} \ell_{ir}^2 d_r, \qquad \ell_{ji} = \frac{m_{ji} - \sum_{r<i} \ell_{jr}\ell_{ir}d_r}{d_i}, \quad \ell_{ii} = 1. \tag{35}$$

All comparisons are exact rational comparisons; no floating-point eigenvalue test is used. Equivalently, each leading principal minor is positive. The accompanying check program reproduces the coefficient identities, kernels and integer comparisons from the tables.

Table 1: Positive lower bounds for the exact successive pivots. In the coercivity rows, $I_o = (0, 1, 2, 4, 5, 6)$.

Matrix	Pivot lower bounds
$B^T S_H B$	$59128, 41179, 936, 64, 193, 142$
$B^T S_G B$	$47368, 1937, 1427, 158, 140, 186$
$(T_H)_{I_o,I_o}$	$3848, 56343, 44452, 368, 162, 270$
$(T_G)_{I_o,I_o}$	$14456, 126713, 32806, 146, 166, 166$
$R^T U_d R$	$19504, 7255, 885$
$R^T V_d R$	$15488, 14526, 589$
U_c	$4336, 11022, 544$
V_c	$7200, 401, 152$

The columns of R span $\mathbf{1}^\perp$. Together with Table 1, the kernel relations show that each of U_d, V_d, S_H and S_G is PSD. The zero future rows give the same conclusion for T_H, T_G, while U_c, V_c are directly positive definite. Every term on the right sides of (32) and (33) is now nonnegative, proving Lemma 1.

4 From the history certificate to weak convergence

The descent estimate and coercivity imply that E_k decreases to a finite nonnegative limit and that both scaled sequences are bounded. They also give the explicit summability estimate

$$\sum_{k=2}^{\infty} \big(\|X_{k+1} - X_k\|^2 + \|Y_{k+1} - Y_k\|^2 + \phi(X_{k+1}) + \psi(Y_{k+1}) \big) \le \frac{E_2}{160}. \tag{36}$$

In particular, the increments tend strongly to zero.

Let $T = \operatorname{diag}(\mathbf{1}_3, \mathbf{1}_3) \in \mathbb{R}^{6\times 2}$. Exact addition of the entries in (14)–(15) gives

$$T^T P T = \begin{pmatrix} 65936 & 0 \\ 0 & 0 \end{pmatrix}, \qquad T^T Q T = \begin{pmatrix} 65936 & 2712 \\ 2712 & 0 \end{pmatrix}. \tag{37}$$

Every component in a three-state history differs from the corresponding current component by a quantity tending strongly to zero. The histories are bounded, so (37) proves

$$E_k - M(X_k, Y_k) \longrightarrow 0, \qquad M(X,Y) = 65936(\|X\|^2 + \|Y\|^2) + 5424\langle KX, Y\rangle. \tag{38}$$

Since $\|K\| \le \sqrt{5/2} < 2$,

$$M(X,Y) \ge 60512(\|X\|^2 + \|Y\|^2). \tag{39}$$

The bounded self-adjoint operator defining M therefore induces an equivalent Hilbert norm. After the fixed scaling (6), this metric is independent of the selected KKT point. Thus the squared distance to every point of Z has a limit in one fixed equivalent metric.

For completeness, consider the product operator

$$\mathcal{A}(x,y) = \big(\partial f(x) + L^* y,\ \partial g^*(y) - Lx\big). \tag{40}$$

It is maximally monotone: the Cartesian product of the subdifferentials is maximally monotone, and the added bounded skew-adjoint linear operator has full domain. Proximal optimality supplies, at (x_{k+1}, y_{k+1}), the residual

$$\left(\frac{x_k - x_{k+1}}{\tau} + L^*(y_{k+1} - y_k), \quad \frac{y_k - y_{k+1}}{\sigma} + \frac{1}{4}L(x_{k+1} - x_k)\right) \in \mathcal{A}(x_{k+1}, y_{k+1}). \tag{41}$$

It tends strongly to zero. Consequently every weak cluster point belongs to Z. One can see the required weak–strong graph closure directly: for every $(v, w) \in \operatorname{gra}\mathcal{A}$, monotonicity gives $\langle z_n - v, r_n - w\rangle \ge 0$. Passing to a weakly convergent subsequence with $r_n \to 0$ gives $\langle \bar{z} - v, -w\rangle \ge 0$ for every such graph point. Maximality implies $0 \in \mathcal{A}\bar{z}$.

The equivalent metric in (38) has the same weak topology. The distance-limit property makes the weak cluster point unique: subtracting the two squared distances to two possible cluster points and passing along the two corresponding subsequences gives the negative of the squared metric distance between them equal to its positive. Hence they coincide. Boundedness and weak sequential compactness in Hilbert space then give weak convergence of the entire sequence. In finite dimensions the same compactness argument is in norm. This proves the quarter-extrapolation part of Theorem 1.

5 The half-extrapolation endpoint

The proof at $\theta = 1/2$ uses the energy already established in [2, Lemma 6]. We include the remaining limiting argument because the last first-increment coefficient in that lemma vanishes at this parameter.

Fix $(x^*, y^*) \in Z$, put $p = \tau\sigma\|L\|^2 < 2$, and write

$$d_k = x_{k+1} - x_k, \qquad e_k = y_{k+1} - y_k, \tag{42}$$

$$F_k = f(x_{k+1}) - f(x^*) + \langle L^* y^*, x_{k+1} - x^* \rangle, \tag{43}$$

$$G_k = g^*(y_{k+1}) - g^*(y^*) - \langle L x^*, y_{k+1} - y^* \rangle. \tag{44}$$

These gaps are finite and nonnegative, starting at $k = 0$. For $L \neq 0$, the hypotheses of the published Lemma 6 permit the literal specialization $\theta = 1/2$, because $p < 2$ is strict. Its energy becomes

$$V_k = \frac{1}{2}F_k + \frac{\|x_{k+1} - x^*\|^2}{2\tau} + \frac{\|y_k - y^* + (\sigma/2)Ld_k\|^2}{2\sigma} + \frac{\|d_k\|^2}{4\tau} - \frac{\sigma}{8}\|Ld_k\|^2, \tag{45}$$

and its inequality is

$$V_{k+1} - V_k + F_k + G_k \leq -\frac{1}{2\sigma}\|e_k + \tfrac{\sigma}{2}L(d_{k+1} - d_k)\|^2 - \frac{2-p}{8\tau}\|d_{k+1} - d_k\|^2. \tag{46}$$

The last two terms of (45) are at least $(2-p)\|d_k\|^2/(8\tau)$. Hence $V_k \geq 0$, both state sequences are bounded, and

$$\sum_{k=0}^{\infty}(F_k + G_k) < \infty, \qquad \sum_{k=0}^{\infty}\|d_{k+1} - d_k\|^2 < \infty, \qquad \sum_{k=0}^{\infty}\|e_k\|^2 < \infty. \tag{47}$$

The last conclusion follows by subtracting the square-summable term $(\sigma/2)L(d_{k+1} - d_k)$ from the other square-summable vector in (46).

Lemma 2. *Let (z_k) be a bounded sequence in a normed space and put $v_k = z_{k+1} - z_k$. If $v_{k+1} - v_k \to 0$, then $v_k \to 0$.*

Proof. For any fixed positive integer m,

$$mv_k = z_{k+m} - z_k - \sum_{j=1}^{m-1}(m-j)(v_{k+j} - v_{k+j-1}). \tag{48}$$

If $\sup_k\|z_k\| \leq C$, taking the limit superior in (48) gives $\limsup_k\|v_k\| \leq 2C/m$. Letting $m \to \infty$ proves the assertion. ∎

Apply Lemma 2 to (x_k). The second conclusion in (47) yields $d_k \to 0$. Together with $F_k \to 0$, boundedness and (45), this shows that the difference between V_k and

$$\frac{\|x_k - x^*\|^2}{2\tau} + \frac{\|y_k - y^*\|^2}{2\sigma} \tag{49}$$

tends to zero. This squared weighted distance therefore has a limit for each KKT point. The residual (41), with 1/4 replaced by 1/2, tends strongly to zero. The graph and distance-limit argument from Section 4 proves weak convergence, and norm convergence in finite dimensions. Notice that this proof does not claim $\sum_k\|d_k\|^2 < \infty$.

The printed source assumes $L \neq 0$. If $L = 0$, the two recurrences decouple into proximal point iterations. Their subgradient inequalities give

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &\leq \|x_k - x^*\|^2 - \|d_k\|^2 - 2\tau F_k, \\ \|y_{k+1} - y^*\|^2 &\leq \|y_k - y^*\|^2 - \|e_k\|^2 - 2\sigma G_k. \end{aligned} \tag{50}$$

They supply boundedness, summability and vanishing residuals directly, and the same graph and weighted-distance argument applies. This establishes the half-extrapolation statement for the entire class of Theorem 1.

The strict product range here is sharp in the universal sense, by the bilinear example from [2, Section 7]. Take $H = G = \mathbb{R}$, $f \equiv 0$, $g^* \equiv 0$, $L = \mathrm{Id}$ and $\tau = \sigma = \sqrt{p}$. The iteration matrix at $\theta = 1/2$ is

$$B_p = \begin{pmatrix} 1 & -\sqrt{p} \\ \sqrt{p} & 1 - 3p/2 \end{pmatrix}. \tag{51}$$

For $p = 2$, the vector $(\sqrt{2}, 2)$ has eigenvalue -1, so it generates a nonconvergent two-cycle. For $p > 2$, the characteristic polynomial evaluated at -1 is $4 - 2p < 0$, and it tends to $+\infty$ on the negative real axis. Thus an eigenvalue lies below -1, and a corresponding nonzero real eigenvector gives divergence in norm. Failure is asserted for these initializations, not for every nonzero initial point.

A Integer coefficient tables

The following tables complete the certificate in Section 3. Rows and columns of the descent weight matrices are ordered $(*, 1, 2, 3)$, and those of the coercivity weight matrices are ordered $(*, 1, 2)$. Diagonal entries are zero. All off-diagonal entries are positive. The norm multipliers U_d, V_d, W_d use state order $(0, 1, 2, 3)$; U_c, V_c, W_c use $(0, 1, 2)$.

$$w^{\phi} = \begin{pmatrix} 0 & 5952 & 73248 & 52672 \\ 128 & 0 & 128 & 90592 \\ 131456 & 84768 & 0 & 128 \\ 128 & 128 & 142976 & 0 \end{pmatrix}. \tag{52}$$

$$w^{\psi} = \begin{pmatrix} 0 & 38360 & 93384 & 128 \\ 90864 & 0 & 152 & 76400 \\ 40720 & 128928 & 0 & 128 \\ 128 & 128 & 76240 & 0 \end{pmatrix}. \tag{53}$$

$$v^{\phi} = \begin{pmatrix} 0 & 128 & 144 \\ 128 & 0 & 50848 \\ 144 & 50848 & 0 \end{pmatrix}. \tag{54}$$

$$v^{\psi} = \begin{pmatrix} 0 & 144 & 3792 \\ 3808 & 0 & 56752 \\ 128 & 60416 & 0 \end{pmatrix}. \tag{55}$$

$$U_d = \begin{pmatrix} 3424 & -8672 & 8208 & -2960 \\ -8672 & 25680 & -29600 & 12592 \\ 8208 & -29600 & 41184 & -19792 \\ -2960 & 12592 & -19792 & 10160 \end{pmatrix}. \tag{56}$$

$$V_d = \begin{pmatrix} 2064 & -2752 & -2736 & 3424 \\ -2752 & 8640 & -9600 & 3712 \\ -2736 & -9600 & 39744 & -27408 \\ 3424 & 3712 & -27408 & 20272 \end{pmatrix}. \tag{57}$$

$$W_d = \begin{pmatrix} 7792 & -2672 & -10624 & 5504 \\ 84256 & 30800 & -130208 & 15280 \\ -104624 & 142272 & 164080 & -70272 \\ 18528 & -97152 & 29424 & 49488 \end{pmatrix}. \tag{58}$$

$$U_c = \begin{pmatrix} 4336 & 464 & -2416 \\ 464 & 11072 & -12016 \\ -2416 & -12016 & 14432 \end{pmatrix}. \tag{59}$$

$$V_c = \begin{pmatrix} 7200 & -6448 & 2480 \\ -6448 & 6176 & -2240 \\ 2480 & -2240 & 1008 \end{pmatrix}. \tag{60}$$

$$W_c = \begin{pmatrix} 10704 & -2752 & -3856 \\ 16672 & -4944 & -33264 \\ -29952 & 9200 & 34704 \end{pmatrix}. \tag{61}$$

All residual matrices are determined by (27)–(30); no unspecified coefficient remains. The finite exact checks establish sufficient inequalities on every actual Hilbert-space trajectory. They do not require any converse interpolation statement or a numerical semidefinite feasibility assumption.

References

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